Homework 4

CALCULUS I (STM 1001)

due: **Friday, February 27, 2026 at 11:30a**

Instructions: Please submit solutions on Populi as a knitted markdown pdf. For handwritten work, please scan your solution pages and include in the final pdf. Homeworks can be completed individually or in groups of up to three people. Names of all group members must be included on the writeups.

Problem 1: Second derivative, concavity, and inflection points

a. Consider $f(x) = 2x^3 - 9x^2 + 12x - 1$.

- Find $f'(x)$ and $f''(x)$ (by hand, using derivative rules).
- Find all critical points (where $f'(x) = 0$). For each, use the second derivative test to classify it as a local maximum, local minimum, or neither.
- Find all inflection points (where $f''(x) = 0$ and concavity changes). State the intervals where f is concave up and where f is concave down.
- In `R` with `mosaicCalc`, define f, compute f' and f'' using `D()`, and plot $f(x)$ over a domain that shows the main features. Add vertical lines (or annotate) at the critical points and inflection points.

b. Consider $g(x) = \dfrac{x}{1+x^2}$.

- Find $g'(x)$ and $g''(x)$.
- Where is g concave up? Where is g concave down? Are there any inflection points?
- In `R`, plot $g(x)$ and $g''(x)$ on the same axes over $[-3, 3]$. Verify your answers from part (b).

Problem 2: Partial derivatives and the gradient

a. Let $f(x, y) = 2x^2y - xy^2 + 4x - 3y$.

- Find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ (treat the other variable as constant when differentiating).
- Write the gradient $\nabla f = \left(\dfrac{\partial f}{\partial x}, \dfrac{\partial f}{\partial y}\right)$.
- Evaluate ∇f at the point $(1, -1)$. In which direction does f increase fastest at that point?

b. Let $h(x, y) = \dfrac{1}{1+x^2+y^2}$.

- Find $\dfrac{\partial h}{\partial x}$ and $\dfrac{\partial h}{\partial y}$. (Use the chain rule: if $h = (1 + x^2 + y^2)^{-1}$, then $\dfrac{\partial h}{\partial x} = -1 \cdot (1 + x^2 + y^2)^{-2} \cdot 2x$.)
- Evaluate ∇h at $(1, 0)$ and at $(0, 0)$. Interpret: where does the gradient point at each of these points?

c. In R, use the `numDeriv` package to check your answers. Define f as a function of a vector `v` where `v[1]` is x and `v[2]` is y, then use `grad(f, c(1, -1))` to compute the gradient at $(1, -1)$. Compare the numerical result to your analytical answer from part (a.iii).

Problem 3: Critical points and the Hessian

a. Let $f(x, y) = 2x^2 + xy + y^2 - 5x - 3y + 2$.

- Find all critical points by solving $\dfrac{\partial f}{\partial x} = 0$ and $\dfrac{\partial f}{\partial y} = 0$ simultaneously.
- Compute the Hessian matrix $H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$.
- For each critical point, compute $D = f_{xx}f_{yy} - (f_{xy})^2$ and use the second derivative test to classify it (local min, local max, or saddle).

b. Let $g(x, y) = x^2 - y^2 + 2xy - 2x + 1$.

- Find all critical points.
- Classify each using the second derivative test.
- In R, use `contour_plot()` from `mosaicCalc` to plot the level curves of g over a domain that includes all critical points. Add points at the critical points. Does the contour plot match your classification?

Problem 4: Portfolio optimization (with R)

In this problem you will work with the two-asset mean-variance portfolio model. Let w be the fraction of wealth invested in asset 1, so $1 - w$ is invested in asset 2. The objective is to maximize

$$f(w) = w\mu_1 + (1 - w)\mu_2 - \lambda(w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\sigma_{12}),$$

where μ_1, μ_2 are expected returns, σ_1^2, σ_2^2 are variances, σ_{12} is the covariance, and $\lambda \geq 0$ is the risk aversion parameter.

a. **By hand:** Suppose $\mu_1 = 0.08$, $\mu_2 = 0.04$, $\sigma_1^2 = 0.09$, $\sigma_2^2 = 0.01$, $\sigma_{12} = 0$.

- Expand $f(w)$ and simplify (expand $(1 - w)^2$ and $2w(1 - w)$, then collect terms).
- Find $f'(w)$.
- Set $f'(w) = 0$ and solve for the optimal w^* when $\lambda = 2$. Does w^* lie in $[0, 1]$?

b. **In R:** Install `tidyquant` if needed (`install.packages("tidyquant")`). Download ETF data for `SPY` and `BND` from 2019-01-01 to the present. Compute daily returns, then annualized mean returns μ_1, μ_2 and covariance matrix (annualized).

- Report your estimated $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \sigma_{12}$.

- For $\lambda = 2$, find the optimal portfolio weights w^* and $1 - w^*$ (you can use `optimize()` or `optim()` in `R`, or solve the first-order condition analytically).
- Plot the efficient frontier: for $\lambda \in \{0.5, 1, 2, 5, 10\}$, compute the optimal w and the resulting portfolio expected return and volatility. Plot volatility (x-axis) vs. expected return (y-axis) for these five portfolios. Add the two individual ETFs (SPY and BND) as points on the same plot.

Problem 5: Gradient descent and a simple neuron

A single neuron with one input computes $\hat{y} = \sigma(wx + b)$ where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid. For one data point (x, y), the MSE loss is $L = \frac{1}{2}(\hat{y} - y)^2$.

a. **Chain rule:** Using the chain rule, show that

$$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}) \cdot x, \qquad \frac{\partial L}{\partial b} = (\hat{y} - y) \cdot \hat{y}(1 - \hat{y}).$$

(Hint: Let $z = wx + b$. Then L depends on $\hat{y}$, $\hat{y}$ depends on z, and z depends on w and b. Apply the chain rule step by step.)

b. **Numerical check:** For $(x, y) = (3, 1)$ and $(w, b) = (0.2, 0)$, compute z, $\hat{y}$, and the gradients $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$ by hand (or with a calculator). Show each step.

c. **In `R`:** Implement gradient descent for this single-neuron model. Use the data point $(x, y) = (3, 1)$, initial $(w, b) = (0.2, 0)$, learning rate $\alpha = 0.5$, and run 20 iterations.

- Report the final (w, b) and the final $\hat{y}$. Did $\hat{y}$ move toward $y = 1$?
- Plot the loss L versus iteration number. Does it decrease?